EART11200: Natural Scientist's Toolkit





This week we will cover the Richter scale, and some atmospheric properties. First, consider logarithms.

Logarithms

- What is a logarithm? Related to exponents.
- It tells us what exponent we need

$$3^{?} = 9$$

Or, how many times we have to multiply 3 by itself to get 9?

Answer: 2 (3x3=9)

$$3^2 = 9$$

$$\log_3(9) = 2$$

Here 3 is called the base, and 2 is the exponent.

'What power is 3 raised to in order to get 9?'

The answer is 2.

Any number can be the base, but the most common base numbers are 10 and e (Euler's number) where e = 2.71828.

$\log_e$ is called the natural logarithm and has the symbol 'ln'

Key point $y = a^x$ and $\log_a y = x$ are equivalent.

WORKED EXAMPLE

20.1 Write down the logarithmic form of the following:

(a) $16 = 4^2$ (b) $8 = 2^3$ (c) $25 = 5^2$

Solution

(a) $16 = 4^2$ may be written as

$$2 = \log_4 16$$

that is, 2 is the logarithm to the base 4 of 16.

(b) $8 = 2^3$ may be expressed as

$$3 = \log_2 8$$

which is read as '3 is the logarithm to the base 2 of 8'.

(c) $25 = 5^2$ may be expressed as

$$2 = \log_5 25$$

that is, 2 is the logarithm to the base 5 of 25.

Key point $\log A + \log B = \log AB$

WORKED EXAMPLES

20.7 Simplify to a single log term

(a) $\log 9 + \log x$
(b) $\log t + \log 4t$
(c) $\log 3x^2 + \log 2x$

Solution

(a) $\log 9 + \log x = \log 9x$

(b) $\log t + \log 4t = \log(t.4t) = \log 4t^2$

(c) $\log 3x^2 + \log 2x = \log(3x^2.2x) = \log 6x^3$

Provided that the base is the same for all terms! This holds for all such relationships between logarithms.

Continued

20.8 Simplify

(a) $\log 7 + \log 3 + \log 2$
(b) $\log 3x + \log x + \log 4x$

Solution

(a) We know $\log 7 + \log 3 = \log(7 \times 3) = \log 21$, and so

$$\log 7 + \log 3 + \log 2 = \log 21 + \log 2$$
$$= \log(21 \times 2) = \log 42$$

(b) We have

$$\log 3x + \log x = \log(3x.x) = \log 3x^2$$

and so

$$\log 3x + \log x + \log 4x = \log 3x^2 + \log 4x$$
$$= \log(3x^2.4x) = \log 12x^3$$

Key point

$$\log A - \log B = \log\left(\frac{A}{B}\right)$$

WORKED EXAMPLES

20.10 Use the second law of logarithms to simplify the following to a single log term:

(a) $\log 20 - \log 10$ (b) $\log 500 - \log 75$ (c) $\log 4x^3 - \log 2x$
(d) $\log 5y^3 - \log y$

Solution

(a) Using the second law of logarithms we have

$$\log 20 - \log 10 = \log\left(\frac{20}{10}\right) = \log 2$$

(b) $$\log 500 - \log 75 = \log\left(\frac{500}{75}\right) = \log\left(\frac{20}{3}\right)$$

(c) $$\log 4x^3 - \log 2x = \log\left(\frac{4x^3}{2x}\right) = \log 2x^2$$

(d) $$\log 5y^3 - \log y = \log\left(\frac{5y^3}{y}\right) = \log 5y^2$$

Continued

20.11 Simplify

(a) $\log 20 + \log 3 - \log 6$
(b) $\log 18 - \log 24 + \log 2$

Solution

(a) Using the first law of logarithms we see that

$$\log 20 + \log 3 = \log 60$$

and so

$$\log 20 + \log 3 - \log 6 = \log 60 - \log 6$$

Using the second law of logarithms we see that

$$\log 60 - \log 6 = \log\left(\frac{60}{6}\right) = \log 10$$

Hence

$$\log 20 + \log 3 - \log 6 = \log 10$$

(b) $$\log 18 - \log 24 + \log 2 = \log\left(\frac{18}{24}\right) + \log 2$$

$$= \log\left(\frac{3}{4}\right) + \log 2$$

$$= \log\left(\frac{3}{4} \times 2\right)$$

$$= \log 1.5$$

Key point $n \log A = \log A^n$

This law applies if n is integer, fractional, positive or negative.

WORKED EXAMPLES

20.14 Write the following as a single logarithmic expression:

(a) $3\log 2$ (b) $2\log 3$ (c) $4\log 3$

Solution

(a) $3\log 2 = \log 2^3 = \log 8$

(b) $2\log 3 = \log 3^2 = \log 9$

(c) $4\log 3 = \log 3^4 = \log 81$

20.15 Write as a single log term

(a) $\frac{1}{2}\log 16$ (b) $-\log 4$ (c) $-2\log 2$ (d) $-\frac{1}{2}\log 0.5$

Solution

(a) $\frac{1}{2}\log 16 = \log 16^{1/2} = \log\sqrt{16} = \log 4$

(b) $-\log 4 = -1.\log 4 = \log 4^{-1} = \log\left(\frac{1}{4}\right) = \log 0.25$

(c) $-2\log 2 = \log 2^{-2} = \log\left(\frac{1}{2^2}\right) = \log\left(\frac{1}{4}\right) = \log 0.25$

(d) $-\frac{1}{2}\log 0.5 = -\frac{1}{2}\log\left(\frac{1}{2}\right) = \log\left(\frac{1}{2}\right)^{-1/2} = \log 2^{1/2} = \log\sqrt{2}$

The constant *e*

- *The constant e=2.71… is a very important constant, which describes many natural processes. So much so it is given its own inverse on a calculator – the natural logarithm.*
- So if we have:

$e^2 = 7.389...$

then

$\ln\left(e^2\right) = \ln\left(7.389...\right)$

which is equal to 2

In general, if:

$e^x = y$

then

$\ln\left(e^x\right) = x$

Key point

The most common exponential expression is

e^x

where e is the exponential constant, 2.71828 . . . .

WORKED EXAMPLE

19.1 Use a scientific calculator to evaluate

(a) e^2 (b) e^3 (c) $e^{1.3}$ (d) $e^{-1.3}$

Solution We use the e^x button giving:

(a) $e^2 = 7.3891$

(b) $e^3 = 20.0855$

(c) $e^{1.3} = 3.6693$

(d) $e^{-1.3} = 0.2725$